

Chapter 1: Introduction to Machine Learning

Machine learning is a subset of artificial intelligence that focuses on building systems that learn from data. It enables computers to improve their performance on a task through experience without being explicitly programmed.

Supervised learning uses labeled training data to learn a mapping function from inputs to outputs. Common algorithms include linear regression, decision trees, support vector machines, and neural networks.

Unsupervised learning finds hidden patterns in unlabeled data. Clustering and dimensionality reduction are common unsupervised techniques.

Reinforcement learning trains agents through rewards and penalties as they interact with an environment. The agent learns a policy to maximize cumulative reward.

Chapter 2: Deep Learning

Deep learning uses artificial neural networks with multiple hidden layers to learn complex representations of data. Convolutional neural networks excel at image recognition while recurrent neural networks handle sequential data.

Transfer learning applies knowledge gained from one task to improve performance on a different but related task, reducing the need for large amounts of training data.

Natural language processing enables computers to understand, interpret, and generate human language. Modern NLP relies heavily on transformer architectures.